

ACICP302ICS MODULE 1 PRACTICAL LAB

Integrated ICS Network Security & Attacks: Analyze, Test & Recover

Final Report

Learner	Wendell Twumasi
Environment	Ubuntu 22.04.5 LTS in VirtualBox
Canonical source	labs/ot-security/student-lab-source/
Git commit	2a66f9bcf253eb3fe31c68ad52794cd535421da8
Assessment scope	Loopback-only OT security simulation

1. Executive Summary

The supplied ACICP302ICS OT security simulation was assessed using the canonical student-lab-source tree. The evaluation established the simulated architecture and normal operational baseline, mapped the Modbus register model, captured and analyzed Modbus/TCP traffic, identified local service exposure, inspected the provided validation guard, and conducted three bounded demonstrations: false-data injection against the direct plant path, slow pressure drift through the guarded path, and replay of an in-range value through the guarded path.

The results indicated a distinct security boundary in the supplied design. The direct Modbus service on 127.0.0.1:5020 is writable without passing through the guard, whereas the guarded path on 127.0.0.1:5021 enforces absolute-range, per-write step, and short-window cumulative-drift checks. The replay test further revealed a specific limitation: the supplied guard failed to detect repeated identical in-range writes.

All demonstrations were conducted against the supplied simulator, with traffic restricted to loopback. We did not execute the archive-original source, use bridged networking or port forwarding, or modify the supplied scripts or target parameters.

2. Scope, Safety and Preflight

Canonical source used: labs/ot-security/student-lab-source/.

Git commit verified: 2a66f9bcf253eb3fe31c68ad52794cd535421da8.

Target/bind addresses remained on 127.0.0.1.

No bridged networking, port forwarding, real ICS equipment, or external target was used.

The archive-original directory was not executed.

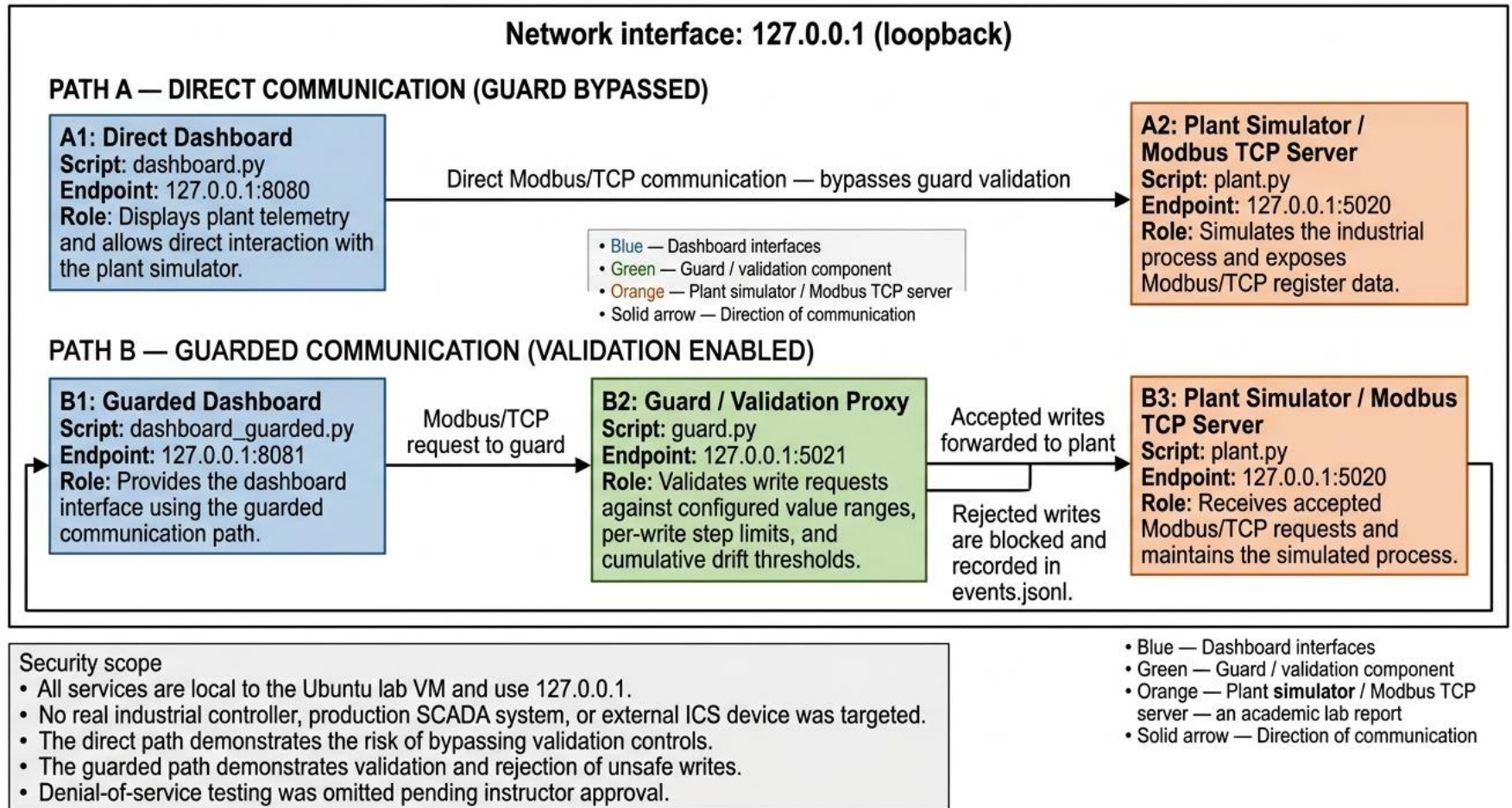
Temporary NAT was used only for approved setup/download activity and was disconnected before isolated lab execution.

The supplied attack scripts were run with their supplied defaults; no target, bind address, request count, duration, or thread-count changes were made.

The DoS demonstration was not performed because the assessment guide requires instructor approval.

3.Architecture and Data Flow

Ubuntu 22.04 Lab VM — Local Loopback Environment



Use:

Figure 1: OT security lab topology showing direct and guarded Modbus/TCP communication paths within the Ubuntu loopback-only environment.

The supplied reference architecture contains four services:

plant.py — simulated process / Modbus-TCP plant service at 127.0.0.1:5020.

guard.py — validation proxy at 127.0.0.1:5021.

dashboard.py — direct dashboard at 127.0.0.1:8080.

dashboard_guarded.py — guarded dashboard at 127.0.0.1:8081.

Direct path: Dashboard 8080 → Plant 5020.

Guarded path: Dashboard 8081 → Guard 5021 → Plant 5020.

The guard is an educational validation layer. It should not be interpreted as a complete production security boundary or process-safety system.

4. Normal Operation and Data Model

The plant simulation exposes four holding registers to Modbus/TCP clients using function code 3 and unit/slave ID 1.

Register	Signal	Raw range	Engineering range	Meaning
0	Temperature	600–750	60.0–75.0 °C	Simulated temperature
1	Pressure	300–400	30.0–40.0 bar	Simulated pressure
2	Flow	500–1500	50.0–150.0 Nm ³ /h	Simulated flow
3	Alarm	0/1	0/1	Alarm state

The initial process values are temperature 670, pressure 340, and flow 900. During normal plant operation, these values may fluctuate slightly within the expected operating range. Temperature, pressure, and flow are monitored against the defined safety

limits, while the alarm register indicates an alarm condition when the pressure reaches or exceeds the configured threshold of 395.

The normal dashboard showed that the process values were reasonable and that the operation was normal, on both the path and the guarded path.

5. Packet Visibility and Modbus Analysis

Packet Analysis Limitation

The captured Modbus/TCP traffic was used to identify the request and correlate it with the corresponding response using the transaction identifier and protocol fields. However, the response decoding did not provide sufficiently reliable evidence to independently confirm the returned register value. Therefore, register values and process-state conclusions are interpreted alongside the documented register map and dashboard evidence rather than inferred from an ambiguous packet decode.

A baseline Wireshark capture was made on the loopback interface. Saved as evidence/pcap/base_modbus_capture.pcapng.

When I filtered the Wireshark capture using `tcp.port == 5020`, I observed Modbus traffic containing request-and-response transactions. This helped me identify the communication between the client and the Modbus/TCP server. However, the capture alone does not reliably confirm the returned register values, so I used the register map and dashboard evidence to support the wider analysis.

6. Local Service Exposure

The supplied simulation exposes local TCP listeners for the plant, guard, direct dashboard, and guarded dashboard. The relevant endpoints are 127.0.0.1:5020, 127.0.0.1:5021, 127.0.0.1:8080, and 127.0.0.1:8081.

The main security observation is that the direct plant Modbus endpoint is a separate path from the guard. A client that can reach the direct endpoint can bypass the validation proxy.

7. Guard Analysis

The supplied guard applies three validation classes:

Absolute range checks: temperature 600 to 750 pressure 300 to 400 and flow 500 to 1500.

Per-write step limits: temperature 30 raw units, which equals 3.0 °C; pressure 50 units, which equals 5.0 bar; flow 100 raw units, which equals 10.0 Nm³/h.

Three-sample cumulative drift checks: temperature threshold 20 raw units or 2.0 °C; pressure threshold 20 raw units or 2.0 bar; flow threshold 40 raw units or 4.0 Nm³/h.

Rejected writes do not get forwarded to the plant. Accepted writes do get. The history gets updated. Rejection events are recorded in events.jsonl.

The guard therefore gives behavioral validation. However its protection relies on traffic that uses the guarded path and on the thresholds being suitable, for the process.

7.1 Pre-Test Plan

All tests were conducted within the supplied Ubuntu OT simulation using loopback-only endpoints. Normal plant telemetry was checked before testing. Evidence was collected from the attack terminal, dashboards, guard logs, and packet capture where applicable.

Test	Script and target	Baseline	Expected outcome	Evidence	Stop Condition	Recovery
False-data injection	plant_sim/false_data_attack.py; 127.0.0.1:5020	Normal plant telemetry	Unsafe values reach the plant through the unguarded path	04A, 04B	Stop after planned iterations or if the simulation becomes unstable	Stop the attack and confirm normal plant telemetry

Slow pressure drift	plant_sim/slow_drift_attack.py; 127.0.0.1:5021	Normal pressure telemetry	Out-of-range or excessive cumulative pressure changes are rejected	05A, 05B, 14	Stop after planned attempts or if the simulation becomes unstable	Stop the test and confirm normal plant telemetry
Replay attack	plant_sim/replay_attack.py; 127.0.0.1:5021	Normal pressure telemetry and existing guard-event log	Check whether repeated identical pressure writes are detected or rejected	06A, 06B	Stop after planned replay attempts or if the simulation becomes unstable	Stop the test and confirm normal plant telemetry

All tests were restricted to the local loopback simulation. Testing was stopped after the planned attempts, and recovery was verified by checking that normal plant telemetry was restored. The replay test also documented a detection limitation because repeated identical pressure writes did not generate new guard events.

8. Controlled Validation Tests

Test	Path	Expected	Observed	Evidence	Recovery
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False-data injection	127.0.0.1:5020	Unsafe values reach plant	110 °C, 55 bar, 200 Nm3/h reached plant; 6/6 unsafe readings	04A, 04B	Normal direct telemetry restored after attack
Slow pressure drift	Guarded 5021	Out-of-range/drift writes rejected	Writes above 40 bar were rejected; cumulative drift events recorded	05A, 05B, 14	Normal telemetry confirmed
Replay	Guarded 5021	Repeated value ideally detected	Three identical 39.4 bar writes produced no new guard event	06A, 06B	Normal telemetry confirmed

9. Validation Test 1 — False Data Injection

I ran the supplied script `false_data_attack.py` against port 5020. I set the script to try a temperature of 110 °C, a pressure of 55 bar and a flow of 200 Nm3/h for six cycles.

What I saw was that the direct plant accepted those values. The terminal showed readings for temperature, pressure and flow in every one of the six cycles and it said that unsafe values had reached the direct plant. The direct dashboard also showed the injected condition. This proves that the direct plant endpoint can bypass the guard.

After I stopped the demonstration the simulated plant went back to telemetry and the direct dashboard once again showed values, inside the normal operating ranges.

10. Validation Test 2 — Slow Drift Through Guard

The `slow_drift_attack.py` script was run against port 5021. It increased pressure by 10 units in each iteration running for 12 iterations total.

The guard responded correctly during the test. It accepted pressure values that stayed within its defined limits. It rejected values that went outside the safe range or showed cumulative drift beyond the allowed threshold.

Event logs recorded pressure readings such as 40.6, 41.6 42.6 and 43.6 bar. These values were above the 40 bar limit meaning they were outside the safe range of 30 to 40 bar. Later attempts also showed drift exceeding the 2.0 bar three-sample threshold.

This shows that the guard is effective. It stops the slow-drift attack from reaching the plant when the guarded path is in use. The guard. Blocks unsafe pressure valve and detects cumulative drift over time.

11. Validation Test 3 — Replay Limitation

The `replay_attack.py` script was run against port 5021. It read the pressure picked a value around 39.4 bar that is within the normal range and sent that same value three times in a row.

What happened: the replay finished with no new guard event being triggered. The event log stayed the same. This shows a weakness in the current guard mechanism. When the same valid pressure value is sent times the guard does not recognize it as a replay attempt.

This result doesn't mean all replay attacks can fool the guard. It only shows that this specific case—repeating the same in-range value—was not caught. The guard failed to detect it because it treats valid inputs as normal even when they are repeated.

12. Risk Register

ID	Risk	Impact	Assessment	Recommended control
R1	Direct Modbus path bypasses guard	High	Demonstrated	Remove unnecessary direct access; enforce authenticated/authorized access and network segmentation.
R2	No demonstrated client authentication/authorization	High	Observed from supplied design	Use appropriate ICS access control, segmentation and authenticated management paths.
R3	Replay of identical in-range values not detected	Medium	Demonstrated limitation	Add sequence/timestamp/context validation and anomaly detection appropriate to the process.
R4	Static thresholds may not represent all process states	Medium	Design limitation	Tune thresholds from validated process requirements and add contextual rules.
R5	Denial-of-service behavior not assessed	Medium	Not performed; approval required	Perform only under approved test conditions with recovery criteria.
R6	Guard has limited process context	Medium	Design limitation	Combine protocol validation with process-aware monitoring and independent safety controls.

13. Recovery and Evidence Handling

After each controlled demonstration, the simulator was allowed to return to normal telemetry and the result was checked through the dashboard/telemetry. The baseline PCAP was copied into evidence/pcap/. The evidence directory structure was created as:

evidence/screenshots/

evidence/pcap/

evidence/report/

The PCAP was verified at approximately 1.2 MB as evidence/pcap/base_modbus_capture.pcapng.

14. Evidence Index

Evidence	Description	Use
01	Wireshark baseline capture	Baseline protocol visibility
02	Normal direct dashboard	Normal operation
03	Normal guarded dashboard	Normal guarded operation
04A	False-data attack terminal	Direct-path validation
04B	Direct dashboard impact	Unsafe-value impact
Recovery 1	Direct dashboard normal after false data	Recovery
05A	Slow-drift attack terminal	Guarded-path validation
05B	Initial slow-drift rejection output	Guard behavior
06A	Replay attack terminal	Replay validation
06B	No new event after replay	Guard limitation

Recovery 3	Normal plant telemetry after replay	Recovery
07	Local Modbus service exposure	Exposure mapping
08	All four local listeners	Architecture/exposure
09	Matching Modbus transaction 2397	Packet analysis
10	Register map/ranges	Data model
11	Direct dashboard baseline	Normal baseline
12	Guarded dashboard baseline	Normal baseline
13	Guard source checks/thresholds	Guard analysis
14	Guard rejection events	Validation evidence

Recovery Verification

Recovery was verified by checking the plant telemetry after the attack tests. The direct dashboard and normal plant telemetry evidence showed that the simulation had returned to normal operating values. Recovery checks were performed after the false-data injection and slow pressure drift tests. The replay test also ended with normal telemetry confirmed. These checks demonstrate recovery of the local simulation; they do not establish recovery of a live industrial control system.

Recovery Evidence

- **Recovery-01-Direct-Dashboard-Normal.png** — evidence of normal readings on the direct dashboard.
- **Recovery-03-Normal-Plant-Telemetry.png** — evidence of normal plant telemetry after testing.

Both screenshots are stored in evidence/screenshots/ in the evidence repository.

15. Evidence-to-Rubric Mapping

Rubric area	Evidence / sections	Coverage
1. Source, scope and architecture	Sections 2–3; commit; Evidence 07–08	Canonical source, loopback scope and architecture established.
2. Data model and normal operation	Section 4; Evidence 02–03, 10–12	Register map, engineering ranges and baseline behavior documented.
3. Packet visibility and exposure	Section 5–6; Evidence 01, 07–09	Modbus/TCP traffic and local service exposure analyzed.
4. Guard analysis	Section 7; Evidence 13–14	Range, step, drift logic and rejection events documented.
5. Validation and recovery	Sections 8–13; Evidence 04–06 and recovery evidence	Three bounded scenarios, observed results and recovery documented.
6. Report and defence	This report; risk register; evidence index	Findings, limitations, risks and traceability consolidated.

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16. Source Traceability

Repository: ACICP302ICS-Network-Security-and-Attacks.

Canonical lab path: labs/ot-security/student-lab-source/.

Verified commit: 2a66f9bcf253eb3fe31c68ad52794cd535421da8.

Relevant source files reviewed: plant.py, guard.py, dashboard.py, dashboard_guarded.py, false_data_attack.py, slow_drift_attack.py, replay_attack.py and supporting documentation.

No source modification was required for the demonstrations documented in this report.

17. Safety Declaration

I confirm that this practical was conducted only against the supplied educational OT simulation in the assigned Ubuntu VM. I did not target real industrial control equipment, external networks, campus/home systems, or another VM. I maintained loopback-only operation, did not use bridged networking or port forwarding, did not execute the archive-original source, and did not alter the supplied attack target parameters. I did not perform the DoS demonstration because instructor approval was required.

Learner name: Wendell Twumasi

Signature:  _____

Date: October 9, 2026

18. Conclusion

The practical established that the supplied simulation provides a useful controlled environment for understanding ICS protocol visibility, direct versus guarded paths, and basic process-value validation. The strongest finding is that the direct Modbus service can be written to without passing through the guard, while the guarded path successfully blocked the tested slow-drift conditions. The replay demonstration exposed a meaningful detection gap because repeated identical in-range writes were not logged as a violation.

The simulation therefore demonstrates both the value and the limitations of simple protocol/process validation. In a production ICS environment, these controls would need to be combined with segmentation, strong access control, monitoring, process-aware anomaly detection, independent safety mechanisms, and carefully approved testing procedures.

19. Submission Checklist

- Final report completed.
- PCAP copied to evidence/pcap/base_modbus_capture.pcapng.
- Evidence subdirectories created.

- Screenshots should be placed in evidence/screenshots/ using the Evidence Index names where possible.
- Final report should be placed in FINAL_REPORT/.
- Before final submission, stop the four lab services and verify that ports 5020, 5021, 8080 and 8081 are no longer listening.
- Keep the learner evidence repository private if required by the assessment brief.